



Irrigation-mediated transfer of geogenic manganese to paddy soils and rice grain in a mine-influenced catchment: a screening-level dietary exposure assessment

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Abstract Manganese (Mn) contamination in agricultural systems is usually attributed to mining or industrial sources, but hydrologically connected catchments can transfer geogenic Mn to staple crops through irrigation. This study characterized Mn transfer from a mine-influenced, recently disturbed upland to irrigation water, stream sediment,

paddy soils, rice tissues, and screening-level dietary exposure in a semi-enclosed catchment in Ilgwang, Busan, South Korea. Surface water and stream sediment were monitored over four agricultural periods in 2021, and paddy soils and rice were sampled from four sequential irrigation zones (P1–P4). Mn was measured by ICP-OES, and solid-phase partitioning was assessed by the modified BCR sequential extraction procedure. Dissolved Mn was strongly enriched along the disturbed-upland drainage pathway (main-flow mean 4.55 ± 1.61 mg/L) relative to tributary inputs (0.16 ± 0.27 mg/L). Stream sediments approached 20,000 mg/kg with more than 97% in non-residual fractions, and paddy soils were enriched up to a field-mean of approximately 1,050 mg/kg (67–83% non-residual). A screening-level, dissolved-Mn flux analysis indicated that the upper paddy zone retained 49.5–56.6% of the seasonal irrigation-derived Mn within the soil–sediment system. Rice grain Mn ranged from 90.64 to 171.47 mg/kg (highest at P3), greatly exceeding typical background levels (1–10 mg/kg). Screening-level hazard quotients exceeded 1 in all zones and remained above 1 for young children even under a conservative 70% bioaccessibility reduction. Because the source was predominantly geogenic and bioaccessibility, cultivar, redox, and pre-disturbance baselines were not directly quantified, these results document an irrigation-mediated exposure pathway rather than a proven causal effect of land disturbance.

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Introduction

Manganese (Mn) is an essential micronutrient for both plants and humans, but excessive exposure can cause adverse neurological and developmental effects, making Mn accumulation in staple foods an important environmental-health concern (Aschner et al., 2007; Erikson et al., 2005). Rice is particularly relevant in this respect because paddy cultivation imposes strong redox oscillations that directly regulate Mn speciation, solubility, and plant availability. Under submerged conditions, Mn(III/IV) oxides can be reduced to the more soluble Mn(II) form, whereas drainage and re-oxidation can regenerate reactive Mn oxides, creating a dynamic Mn shuttle between soil solids and porewater (Huang & Zhang, 2020; Kirk, 2004; Wang et al., 2022). Rice Mn uptake is further shaped by membrane transport and genotype-dependent partitioning, including transporters such as OsN-RAMP5 and loci associated with grain Mn accumulation (Ishimaru et al., 2012; Liu et al., 2017; Xu et al., 2023).

Although Mn contamination in rice has been documented in mining- and groundwater-affected environments, many previous studies have focused on pot experiments, genotype screening, or broad regional surveys rather than on field-scale pathways linking an upland source to irrigation water, paddy soils, and grain exposure (Jaafar et al., 2021; Pinson et al., 2015; Wang et al., 2022). This distinction matters because Mn transfer in agricultural landscapes depends not only on source strength but also on hydrological connectivity, seasonal flooding, and soil redox dynamics (Kirk, 2004; Wang et al., 2022).

Land disturbance can intensify these processes by exposing fresh mineral surfaces, altering runoff pathways, and increasing water–rock interaction. Golf-course development is relevant in this context because construction typically involves extensive grading, excavation, drainage engineering, and sustained irrigation, all of which can modify downstream water chemistry and strengthen source-to-receptor connectivity (Cai et al., 2002; Winter & Dillon, 2005). This mechanism is particularly important in

mine-influenced terrains where excavation can expose sulfide-bearing and Mn-bearing materials to oxygenated water, thereby accelerating acid rock drainage (ARD)-like weathering and metal release. In the Ilgwang district, mineralized granitoids and hydrothermal breccia systems contain sulfide minerals such as arsenopyrite (FeAsS), pyrrhotite (Fe_{1-x}S), and chalcopyrite (CuFeS₂), providing a plausible geogenic source of Fe, Mn, and associated trace metals when weathered materials are newly exposed (Fletcher, 1977). Mn is hosted mainly by Fe–Mg silicates (biotite, chlorite) and secondary Mn-oxide phases in this granitoid–hydrothermal system (Gilkes & McKenzie, 1988).

The Busan–Ilgwang district provides a plausible setting for such a mechanism. Previous studies from nearby pyrophyllite and abandoned mine sites documented acidic, metal-rich drainage and secondary precipitates associated with mine-drainage evolution in the region (Kang et al., 2010; Park et al., 2000). More recent studies from the Ilgwang mine further showed that the drainage remains strongly contaminated and that treatment or neutralization generates substantial precipitates while removing Fe, Al, Mn, Cu, and Zn from solution (Cheong et al., 2025; Lee & Yim, 2023). This regional mine-drainage context is especially relevant to Mn because Fe is commonly removed earlier through oxidation and precipitation as Fe-(oxy) hydroxides, whereas Mn can remain relatively mobile over a wider pH range and may persist farther downstream unless alkalinity becomes sufficiently high or coprecipitation/sorption with Fe- and Al-rich phases becomes effective (Cheong et al., 2025; Woo & Choi, 2001). Accordingly, visible orange to reddish precipitates near the source do not necessarily imply efficient Mn removal from irrigation-relevant flow paths.

In the present catchment, runoff from an upstream golf course enters a tributary that supplies irrigation water to downstream rice paddies, and reddish to orange precipitates have been observed along the drainage pathway. This setting offers an opportunity to examine whether earthworks in a mine-influenced catchment can accelerate the mobilization of predominantly geogenic Mn and establish an irrigation-mediated transfer pathway to paddy soils and rice grain (Blowes et al., 2003; Nordstrom, 2011). Once introduced into paddies, Mn may be retained initially by settling or sorption, but can later be redistributed

under flooded, reducing conditions. Thus, hydrological connection between a disturbed upland source and downstream paddies may create a repeated cycle of Mn mobilization, retention, and remobilization that increases the probability of crop uptake.

Within this framework, we examined whether an irrigation-connected, mine-influenced catchment transfers reactive, predominantly geogenic Mn to paddy soils and rice grain at concentrations relevant to dietary exposure. Recent land disturbance (golf-course construction) is considered a plausible factor that may have enhanced Mn mobilization through ARD-/AMD-like weathering; however, in the absence of pre-disturbance baseline and reference-catchment data, this study documents the source-to-receptor transfer pathway and the associated exposure rather than testing land disturbance as a cause. To this end, we quantified Mn concentrations in surface water, stream sediments, paddy soils, and rice tissues along a connected flow path, evaluated Mn partitioning using sequential extraction, performed a screening-level flux partitioning and topsoil inventory analysis, and assessed screening-level dietary exposure associated with rice consumption.

Materials and methods

Study area and regional geochemical setting

The study area is a semi-enclosed catchment located in Ilgwang, Busan, South Korea. A single hydrologically connected flow path links an engineered outflow structure at an upstream golf course to a tributary stream and downstream rice paddies. The paddy fields are subdivided into sequential irrigation zones along the flow path, allowing the spatial tracking of Mn across surface water, stream sediment, paddy soil, and rice tissues (Fig. 1a).

The catchment is underlain by fractured biotite granite and schist and lies within the broader Busan–Ilgwang mining district, where historical extraction of kaolin, pyrophyllite, and tungsten has been reported. The broader Ilgwang mine district includes mineralized granitoids and hydrothermal breccia systems in which sulfides and metal-bearing alteration minerals occur, providing a plausible geogenic source for Fe, Mn, and associated trace metals when weathered materials are newly exposed

(Fletcher, 1977; Yang & Bodnar, 2004). Previous field studies from the abandoned Ilgwang mine reported highly acidic drainage ($\text{pH} < 3$) under oxidizing conditions (Eh 400–600 mV), with dissolved metals released in the order $\text{Fe} > \text{Cu} > \text{Zn} > \text{Mn} > \text{As} > \text{Cd}$, indicating that Mn is a persistent component of the regional drainage chemistry (Kang et al., 2010; Park et al., 2000). More recent studies further showed that Ilgwang mine drainage remains chemically reactive and produces substantial metal-rich precipitates during treatment or lime neutralization (Cheong et al., 2025; Lee & Yim, 2023). This regional context indicates that the study area is geochemically vulnerable to disturbance-enhanced mobilization of redox-sensitive metals.

In 2018, a golf course was constructed in the upper part of the catchment, involving excavation, terrain modification, and engineered drainage that exposed weathered bedrock and altered natural runoff pathways. In this context, the golf-course construction is interpreted not primarily as a direct chemical source, but as a land-disturbance event that likely enhanced water–rock interaction, exposed reactive materials, and increased hydrological connectivity between the upland source area and downstream agricultural land. Reddish to orange precipitates observed in the downstream drainage channel suggest the formation of Fe- and Mn-rich secondary oxides along the flow path (Fig. 1b). Because runoff from the golf-course outflow is used directly for irrigation in downstream paddies, the catchment provides a field-scale setting for evaluating whether land disturbance in a mine-influenced terrain can establish an irrigation-mediated Mn transfer pathway to agricultural soils and rice grain.

Sampling design and hydrological zoning

Sampling was designed to follow Mn transport along a hydrologically connected pathway from the golf-course effluent to downstream rice paddies. Surface water and streambed sediment were collected at seven sites (S1–S7) distributed along the main flow path from the effluent outlet to the downstream reach. Three additional surface-water sites (S11–S13) were included to characterize intersecting tributary inputs and to distinguish the main Mn transfer pathway from lateral inflows (Fig. 1a).

Field campaigns were conducted in March, May, July, and September 2021, corresponding to

Fig. 1 **a** Study area and hydrological sampling design in the Ilgwang agricultural catchment, Busan, South Korea. The figure shows the golf-course effluent outlet, main-path surface-water and stream-sediment sampling sites (S1–S7), tributary-input surface-water sites (S11–S13), paddy-soil and rice sampling zones (P1–P4), irrigation channels (IC1–IC5), and **b** an inset photograph of reddish-orange Fe/Mn-rich precipitates observed along the downstream drainage pathway



the fallow, sowing, growing, and harvest periods, respectively. Paddy soils and rice plants were sampled from four irrigation zones (P1–P4) arranged sequentially along the irrigation pathway. In total, 92 environmental samples were collected during the study period. Site descriptions, hydrological roles, and sampling matrices are summarized in Table 1.

Sample collection and pretreatment

Surface-water samples were collected during each field campaign to assess spatial and seasonal variation in Mn concentrations and the influence of irrigation activity on downstream transport. At each site, in situ pH and electrical conductivity were measured using

Table 1 Hydrological description of sampling locations used to track Mn transfer from the golf-course effluent to downstream paddies

Site	Distance from effluent outlet (m)	Sampling matrix	Remarks
S1	0	Water, sediment	Golf-course effluent outlet
S2	35	Water, sediment	Upstream of IC1
S3	55	Water, sediment	After IC1; within/adjacent to paddy area
S4	80	Water, sediment	Between IC2 and IC3
S5	100	Water, sediment	After IC3
S6	115	Water, sediment	Between IC4 and IC5
S7	125	Water, sediment	Downstream of all irrigation channels
S11	—	Water	Intersecting tributary 1
S12	—	Water	Intersecting tributary 2
S13	—	Water	Intersecting tributary 3

IC, irrigation channel;

a portable multiparameter meter (YSI Professional Plus, Yellow Springs, OH, USA). Surface-water samples were collected in pre-cleaned bottles and filtered through 0.45 μm membrane filters. Samples were acidified with ultrapure HNO_3 to $\text{pH} < 2$ and stored at 4 $^\circ\text{C}$ until analysis.

Two types of solid-phase samples were collected: streambed sediment and paddy soil. Streambed sediment represents depositional material accumulated within the stream channel and reflects Mn transport and deposition along the water flow path. Sediment samples were collected during the four field campaigns from the upper 5 cm of the streambed using a stainless-steel grab sampler at sites S1–S7. Paddy soils were collected from actively cultivated rice fields located downstream of the stream. Soil samples were obtained from the 0–15 cm root zone in July and September. At each irrigation zone, one composite sample was prepared by mixing subsamples from three points to account for within-zone spatial variability. Both sediment and soil samples were air-dried, ground, and passed through a 2 mm sieve for homogenization. For total Mn analysis, a subsample was further sieved to $< 150 \mu\text{m}$.

Rice plants were collected from each irrigation zone during two stages of the cultivation period: July (growth stage) and September (harvest stage). At each zone, three representative plants were collected. All sampled paddies were planted with the japonica rice cultivar ‘Sugwang’, a mid-late maturing variety adapted to the southern coastal region of Korea (RDA 2010). July samples were separated into roots and shoots, whereas September harvest samples were separated into roots, shoots, and grain. Plant tissues

were rinsed with deionized water to remove adhering soil particles, oven-dried at 80 $^\circ\text{C}$, cut with ceramic scissors, and ground to a 40-mesh fine powder using an agate mortar and an agate mill to avoid trace-metal contamination prior to digestion.

Chemical analysis and sequential extraction

All Mn analyses were conducted at the Earth and Environmental Convergence Analysis Center at Pukyong National University using inductively coupled plasma optical emission spectrometry (ICP-OES; PerkinElmer Optima 7000 DV, PerkinElmer, Waltham, MA, USA). Filtered water samples were analyzed for dissolved Mn. Total Mn in sediment and soil was determined after acid digestion of the $< 150 \mu\text{m}$ fraction, and plant-tissue digests were analyzed following acid digestion of the dried powder.

For plant-tissue digestion, approximately 0.5 g of each powdered sample was treated with a mixture of concentrated sulfuric acid and perchloric acid and heated at 310 $^\circ\text{C}$ until a clear solution was obtained.

Chemical partitioning of Mn was evaluated using the modified BCR sequential extraction procedure (Rauret et al., 1999). Sequential extraction was conducted on stream sediment and selected paddy-soil samples to operationally separate Mn into four fractions: acid-soluble/exchangeable, reducible, oxidizable, and residual. These fractions were used to evaluate Mn lability and the relative importance of potentially mobile Mn pools in the stream–paddy continuum.

Quality assurance/quality control

Quality assurance/quality control (QA/QC) included certified reference materials (CRMs), procedural blanks, and triplicate instrumental measurements. The CRMs used were SLRS-6 (National Research Council Canada; certified Mn = 2.12 ± 0.10 µg/L) for water, SSAR-1 (U.S. Geological Survey) for sediment and soil, and TORT-3 (National Research Council Canada) for biological tissue digests. These CRMs were analyzed concurrently with field samples to verify analytical accuracy.

Measured recoveries for Mn ranged from 94 to 107%, and relative standard deviations were below 5% across sample types. Procedural blanks were included to monitor background contamination during digestion and analysis. Recovery of the summed BCR fractions relative to an independently measured pseudo-total Mn concentration (aqua regia digestion of a parallel subsample) ranged from 92–108%. A matrix-specific QA/QC summary, including certified and measured values for each CRM, is provided in Table S1.

All measurements were performed in triplicate to assess analytical precision, and relative standard deviations were consistently below 5% across sample types, indicating reliable quantification.

Statistical analysis

Results are presented as mean $\pm$ standard deviation. Surface-water data were summarized by sampling site and campaign, whereas soil and plant data were summarized by irrigation zone and sampling period. Ratios of mobile Mn fractions to the residual fraction were calculated to compare Mn lability between stream sediment and paddy soil.

Differences among sites or irrigation zones were tested using one-way ANOVA followed by Tukey's HSD test after checking normality and homogeneity of variance. When assumptions were not met, the Kruskal–Wallis test followed by Dunn's multiple comparison test was applied. All data processing and figure preparation were performed using Python (version 3.12), utilizing libraries such as Pandas for data manipulation, SciPy and Statsmodels for statistical analysis, and Matplotlib and Seaborn for visualization.

Screening-level Mn flux partitioning and topsoil inventory scenarios

To evaluate the relative magnitude of Mn transfer, crop accumulation, and apparent retention within the paddy system, a screening-level Mn flux analysis was conducted for the upper irrigation zone (P1). This zone was selected because it receives the earliest and most direct contact with Mn-enriched irrigation inflow and exhibited the highest paddy-soil Mn concentration among the four paddy zones.

The seasonal Mn input flux through irrigation water was estimated as:

$$F_{\text{in}} = C_{\text{in}} \times D_{\text{in}} \quad (1)$$

where F_{in} is the seasonal Mn input flux (mg/m²/season), C_{in} is the representative Mn concentration of irrigation water entering P1 (mg/L), and D_{in} is the seasonal irrigation depth (L/m²/season). Because 1 mm of irrigation depth is equivalent to 1 L/m², irrigation depth in millimeters or liters per square meter was used directly in the flux calculation. Seasonal irrigation depth was treated as a scenario parameter with low, central, and high values of 800, 1000, and 1200 L/m²/season, respectively, to represent a plausible range for irrigated paddy systems under flooded to water-saving management conditions (Bwire et al., 2024; Zhao et al., 2023).

The seasonal Mn outlet flux from P1 was estimated as:

$$F_{\text{out}} = C_{\text{out}} \times \alpha \times D_{\text{in}} \quad (2)$$

where F_{out} is the Mn flux exported from P1 (mg/m²/season), C_{out} is the representative Mn concentration of water leaving the upper paddy reach (mg/L), and α is the fraction of incoming irrigation water discharged from the paddy as outlet flow. Because direct measurements of outlet discharge were unavailable, α was treated as a hydrological sensitivity parameter rather than a fixed field measurement, and low, central, and high values of 0.6, 0.7, and 0.8 were used.

Whole-plant Mn accumulation was estimated from harvest-stage dry biomass and tissue Mn concentrations as:

$$F_{\text{plant}} = (C_{\text{root}} \times B_{\text{root}}) + (C_{\text{shoot}} \times B_{\text{shoot}}) + (C_{\text{grain}} \times B_{\text{grain}}) \quad (3)$$

where C_{root} , C_{shoot} , and C_{grain} are Mn concentrations in root, shoot, and grain tissues (mg/kg), and B_{root} , B_{shoot} , and B_{grain} are the corresponding dry biomasses per unit area (kg/m²).

The apparent retained Mn flux was calculated as:

$$F_{\text{retained,app}} = F_{\text{in}} - F_{\text{out}} - F_{\text{plant}} \quad (4)$$

This term represents the fraction of incoming Mn not recovered in outlet water or plant biomass and was interpreted as an apparent non-plant flux associated with soil and sediment retention and/or redistribution through percolation, seepage, drainage, and lateral flow.

For comparison with the seasonal flux terms, the excess Mn inventory in topsoil was estimated as:

$$I_{\text{soil,excess}} = (C_{\text{soil}} - C_{\text{ref}}) \times \rho_b \times d_a \quad (5)$$

where $I_{\text{soil,excess}}$ is the excess Mn inventory in topsoil (mg/m²), C_{soil} is the measured topsoil Mn concentration (mg/kg), C_{ref} is the reference soil Mn concentration used to represent background conditions (mg/kg), ρ_b is bulk density (kg/m³), and d_a is the assumed active Mn-enriched soil depth (m). The active Mn-enriched depth was varied as 0.05, 0.10, and 0.15 m to reflect the possibility that Mn accumulation was concentrated in the uppermost portion of the plow layer rather than uniformly distributed through the entire sampled depth. These values were selected with reference to reported plow-layer depths in paddy soils, which commonly approach 0.20 m but may vary substantially among fields and management histories (Hur et al., 2023). Bulk density was set to 1.41 g/cm³, representing the characteristic value for the study area's soil as provided by the national soil database (RDA 2010).

To evaluate whether the present topsoil Mn inventory could plausibly be explained by recent dissolved Mn loading, a back-calculated mean inflow concentration was estimated as:

$$C_{\text{in,req}} = \frac{I_{\text{soil,excess}}/N + F_{\text{plant}} + F_{\text{out}}}{D_{\text{in}}} \quad (6)$$

where $C_{\text{in,req}}$ is the mean inflow Mn concentration required to account for the observed excess topsoil inventory under the assumed contamination duration N , expressed in cultivation seasons. The

contamination duration was varied as 2, 3, and 4 seasons to represent plausible post-construction loading periods.

Reference soil Mn concentration was treated as an uncertainty parameter because an uncontaminated local reference soil was unavailable. Low, central, and high values of 500, 700, and 1000 mg/kg were used to bracket commonly reported background-to-geogenic Mn levels in soils and parent materials, including average upper continental crust abundance and the possibility of elevated lithogenic background in Mn-bearing terrains (Bourgault et al., 2022; Gilkes & McKenzie, 1988).

Scenario ranges for irrigation depth, outlet-flow fraction, contamination duration, active Mn-enriched soil depth, and reference soil Mn concentration are summarized in Table 2. Because the present analysis does not explicitly include all hydrological and geochemical pathways, the resulting values were interpreted as screening-level Mn flux and inventory estimates rather than as a closed mass balance. The outlet-flow fraction was treated as a hydrological sensitivity parameter because direct outlet discharge measurements were unavailable, whereas irrigation depth, active soil depth, and reference soil Mn concentration were constrained using literature-based scenario ranges.

Screening-level dietary risk assessment

Dietary exposure to Mn through rice consumption was evaluated as a screening-level, non-carcinogenic risk assessment based on total Mn concentrations measured in rice grain. Four age groups were considered: children (0–5 years), adolescents (6–18 years), adults (19–65 years), and elderly (≥ 66 years). Age-specific rice intake rates (IR) and body weights (BW) were obtained from the Korea Health Industry Development Institute (KHIDI 2023), and three exposure-frequency scenarios were considered: daily consumption (360 days/year), moderate consumption (180 days/year), and occasional consumption (90 days/year). The exposure parameters used in the calculations are summarized in Table 3.

Estimated daily intake (EDI, mg/kg/day) was calculated as:

$$EDI = \frac{C \times IR \times EF}{BW \times 365} \quad (7)$$

Table 2 Scenario ranges and measured values used for screening-level Mn flux partitioning and topsoil inventory analysis in the upper paddy zone (P1)

Category	Parameter	Symbol	Value or range	Unit	Basis
Measured field values	Paddy area	A	893.37	m ²	Measured
	Inlet Mn con	C _{in}	4.22	mg/L	Measured mean
	Outlet Mn con	C _{out}	0.61	mg/L	Measured mean
	Topsoil Mn con	C _{soil}	1690.04	mg/kg	Measured mean
	Root Mn con	C _{root}	1246.73	mg/kg	Measured mean
	Shoot Mn con	C _{shoot}	861.60	mg/kg	Measured mean
	Grain Mn con	C _{grain}	124.40	mg/kg	Measured mean
	Root dry biomass	B _{root}	0.33	kg/m ²	Measured mean
	Shoot dry biomass	B _{shoot}	1.12	kg/m ²	Measured mean
	Grain dry biomass	B _{grain}	1.32	kg/m ²	Measured mean
	Bulk density	ρ _b	1.41	g/cm ³	Reference value
Scenario parameters	Seasonal irrigation depth	D _{in}	800 / 1000 / 1200	L/m ² /season	Low / central / high scenario
	Outlet-flow fraction	α	0.60 / 0.70 / 0.80	–	Hydrological sensitivity scenario
	Contamination duration	N	2 / 3 / 4	seasons	Post-construction scenario
	Active Mn-enriched soil depth	d _a	0.05 / 0.10 / 0.15	m	Surface-enrichment scenario
	Reference soil Mn con	C _{ref}	500 / 700 / 1000	mg/kg	Background scenario

Seasonal irrigation depth was treated as a literature-based scenario parameter representative of irrigated paddy systems (Bwire et al., 2024; Zhao et al., 2023). Active Mn-enriched soil depth was varied to reflect possible surface-focused accumulation within the plow layer of paddy soils (Hur et al., 2023). Reference soil Mn concentration was varied to bracket plausible background-to-geogenic Mn levels in soils and parent materials (Bourgault et al., 2022; Gilkes & McKenzie, 1988). The outlet-flow fraction was treated as a hydrological sensitivity parameter because direct outlet discharge measurements were unavailable

Table 3 Age-specific exposure parameters used for screening-level Mn intake calculations from rice consumption

Age group	Rice intake, IR (g/person/day)	Body weight, BW (kg)	Exposure-frequency scenarios, EF (days/year)
Children (0–5)	112.45	15.53	90, 180, 360
Adolescents (6–18)	139.69	46.85	90, 180, 360
Adults (19–65)	133.76	67.57	90, 180, 360
Elderly (≥ 66)	150.69	60.61	90, 180, 360

IR and BW were obtained from KHIDI (2023). EF scenarios represent occasional, moderate, and daily rice consumption

where *C* is the Mn concentration in rice grain (mg/kg), *IR* is the daily rice intake (g/person/day), *EF* is the exposure frequency (days/year), and *BW* is body weight (kg).

The hazard quotient (HQ) was calculated as:

$$HQ = \frac{EDI}{RfD} \quad (8)$$

where *RfD* represents the oral reference dose for Mn (0.14 mg/kg/day) (US EPA 2022; US EPA 2004). It indicates the maximum daily Mn intake (mg/kg/day)

that is not expected to cause adverse health effects over a lifetime. When the HQ value is below 1, Mn exposure is considered to be within a safe range, suggesting a low likelihood of negative health effects. However, if HQ is equal to or greater than 1, there is a potential risk of non-carcinogenic health effects due to excessive Mn intake.

Table 4 Summary of pH, electrical conductivity, and dissolved Mn concentrations in surface water collected from the main flow path (S1–S7) and tributary inputs (S11–S13) in March 2021

Site group	pH	Conductivity (mS/cm)	Mn (mg/L)
S1–S7	6.87 ± 0.37	0.30 ± 0.08	4.55 ± 1.61
S11–S13	7.92 ± 0.66	0.22 ± 0.02	0.16 ± 0.27

Dissolved Mn represents the operationally defined 0.45 μm -filtered fraction and does not by itself confirm oxidation state. Because in situ Eh was not measured, the assumption that this fraction is predominantly Mn^{2+} is supported indirectly by the slow abiotic oxidation kinetics of Mn^{2+} by O_2 at the measured circumneutral pH (Diem & Stumm, 1984; Morgan, 2005). Values are presented as mean $\pm$ SD ($n=3$)

Results and discussion

Mn enrichment in source water and irrigation-path attenuation

Table 4 summarizes the pH, electrical conductivity, and dissolved Mn concentrations of surface water collected in March 2021, whereas the full surface-water chemistry dataset is provided in Supplementary Table S2. Across the study area, pH and electrical conductivity showed relatively limited variation, and the concentrations of other measured trace elements remained low overall. In contrast, Mn was distinctly enriched along the main hydrological flow path. The mean dissolved Mn concentration at the main-flow sites (S1–S7) was 4.55 ± 1.61 mg/L, whereas the tributary-input sites (S11–S13) showed a much lower mean concentration of 0.16 ± 0.27 mg/L. These results indicate that Mn enrichment was strongly associated with the golf-course-connected drainage pathway rather than with the surrounding tributary system.

The spatial and seasonal pattern shown in Fig. 2 further supports this interpretation. Dissolved Mn concentrations varied markedly with both distance from the effluent outlet and sampling season, with peak concentrations exceeding 6 mg/L at the upstream sites (S1–S2). These concentrations were far above typical background levels reported for surface waters and exceeded previously reported regulatory or advisory benchmarks for Mn in water, indicating severe Mn enrichment in the source-water pathway (Mandal et al., 2019; McMahon et al., 2019). Regardless of season, Mn concentrations generally

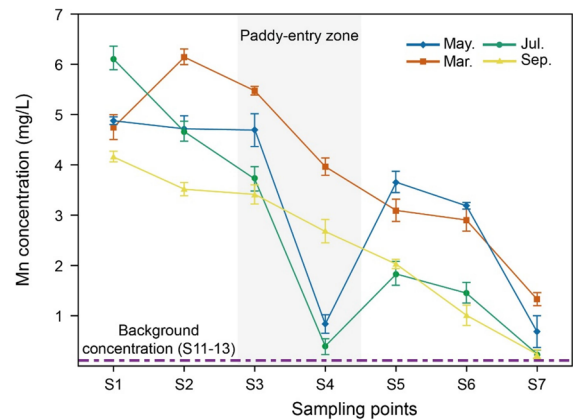


Fig. 2 Spatial and seasonal variation of dissolved Mn concentrations along the main hydrological flow path (S1–S7). Concentrations are shown for March, May, July, and September 2021. The shaded band indicates the paddy-entry attenuation zone (S3–S4), where dissolved Mn declined sharply during the irrigation months. The dashed horizontal line represents the mean background Mn concentration derived from tributary-input sites (S11–S13). Values are presented as mean $\pm$ SD ($n=3$) based on three samples at each site

declined downstream, and the sharpest attenuation occurred across the paddy-entry reach (S3–S4) during the irrigation months, particularly in May and July. In these months, Mn concentrations decreased by more than 70% between S2 and the paddy-entry reach, whereas the downstream decline was less abrupt in March and September, when irrigation was absent or greatly reduced. The dashed line in Fig. 2 represents the mean background Mn concentration derived from the tributary-input sites (S11–S13), showing that Mn concentrations along the main flow path remained substantially above background throughout the monitoring period.

The rapid decline in dissolved Mn at the paddy-entry reach is unlikely to reflect simple downstream dilution alone. Rather, it is more consistent with the combined influence of hydrological redistribution and geochemical removal processes operating as Mn-rich water entered the irrigated paddy system. Flow deceleration at the paddy margin likely promoted settling of Mn-rich particulates, while contact with wet soil surfaces may have enhanced sorption and localized accumulation. In addition, Mn behavior in wetland and paddy environments is strongly controlled by redox dynamics, and transitions in oxygen status can rapidly shift Mn between

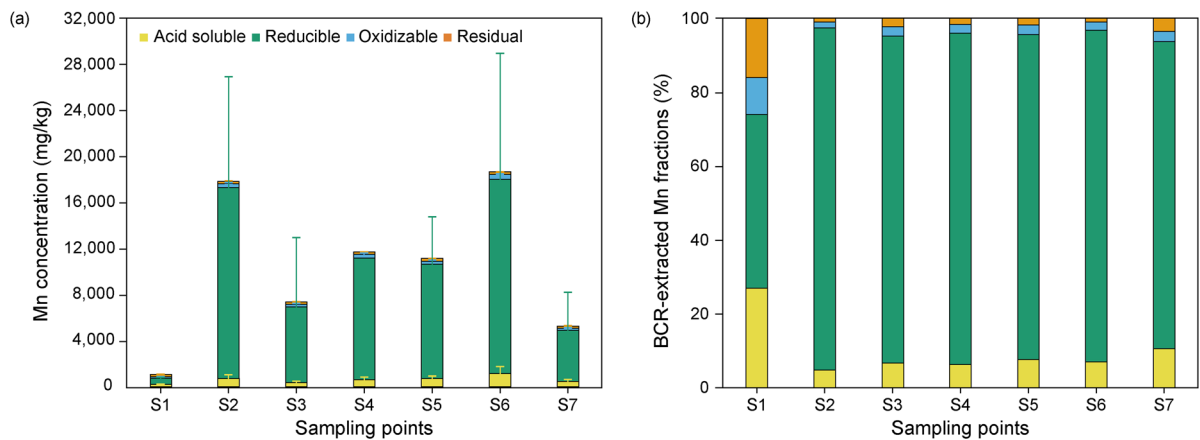


Fig. 3 Mn concentrations and operationally defined fractions in stream sediments collected along the main hydrological flow path. **a** Total Mn concentrations in stream sediments at S1–S7, partitioned into acid-soluble, reducible, oxidizable, and residual fractions using the modified BCR sequential extraction procedure. **b** Relative proportions of BCR-extracted Mn frac-

tions at each sampling point, showing the strong dominance of non-residual Mn pools, particularly the reducible fraction, in most downstream sediments. Values are based on three samples per site; panel **a** shows mean $\pm$ SD ($n=3$), whereas panel **b** shows the mean fractional composition

dissolved and solid-phase forms (Huang & Zhang, 2020; Wang et al., 2022). Accordingly, the paddy-entry zone appears to function as an interception and transformation interface between the disturbed upland source area and the cultivated fields.

Taken together, the surface-water data establish a clear source-to-entry signal: Mn-rich water was generated upstream, transported along the engineered flow path, and sharply attenuated once it encountered the paddy irrigation system. This hydrological pattern provides the basis for understanding the downstream accumulation of Mn in stream sediments, paddy soils, and rice tissues. The observed surface-water pattern is consistent with the geochemical separation of Fe and Mn commonly documented in mine-drainage systems. Under acidic and oxidizing conditions, Fe released from sulfide weathering is readily hydrolyzed and precipitated as Fe-(oxy)hydroxides, whereas Mn often remains more soluble and can therefore persist farther downstream unless pH increases sufficiently or favorable coprecipitation and sorption reactions occur (Cheong et al., 2025; Kang et al., 2010; Woo & Choi, 2001). This behavior provides a plausible explanation for the coexistence of visible orange precipitates near the source and continued Mn transport toward the paddy-entry reach.

Stream sediments as a reactive Mn reservoir

The stream-sediment results show that Mn was not only transported in dissolved form, but was also efficiently stored in the channel as a highly reactive solid-phase reservoir (Fig. 3). Mn concentrations were strongly elevated throughout the main flow path, with the highest concentrations observed at S2 and S6, both approaching 20,000 mg/kg, whereas S1 showed concentrations below 2,000 mg/kg. Downstream of S2, concentrations decreased at S3–S5 to approximately 8,000–11,000 mg/kg, but increased again at S6 to values comparable to those at S2. This non-linear pattern, particularly the re-emergence of a major hotspot at S6 after the decline observed at S3–S5, indicates that sedimentary Mn accumulation was governed not by progressive downstream dilution alone but by localized hydrological and geomorphic controls on deposition and remobilization.

As shown in Fig. 1, this pattern is consistent with the terraced morphology of the channel and the longitudinal variation in flow energy along the stream. The relatively flat reaches near S2 and S6 likely promoted temporary flow stagnation and deposition of Mn-rich particles, whereas the intermediate sections between S3 and S5 were more conducive to active transport. In this sense, the stream functioned not only as a conveyance pathway but also as a depositional sink and

recycling compartment for Mn-rich material. This interpretation is important because it links the high dissolved Mn observed in the source-water pathway to a downstream particulate reservoir capable of sustaining continued Mn release.

Sequential extraction further demonstrates that the sedimentary Mn pool was overwhelmingly labile (Fig. 3b). More than 97% of total Mn occurred in non-residual fractions, and more than 95% was associated with the acid-soluble and reducible fractions. Among the non-residual pools, the reducible fraction dominated at most sites, indicating that Mn was largely associated with redox-sensitive oxide phases rather than with stable mineral lattices. In contrast to the downstream sites, S1 showed a relatively larger acid-soluble proportion, suggesting that Mn speciation at the immediate outlet differed from that at the downstream depositional hotspots. These results indicate that sedimentary Mn was hosted predominantly by geochemically reactive phases, consistent with the reducible BCR fraction being operationally designed to target Fe–Mn (oxyhydr)oxide phases (Tessier et al., 1979), and with XRD-based studies of mine-drainage-derived Mn precipitates elsewhere, which commonly identify poorly crystalline birnessite and todorokite as the dominant secondary Mn-oxide phases under comparable redox and pH conditions (Post, 1999; Tan et al., 2010).

However, a purely anthropogenic interpretation would overstate the implication of these data by suggesting that the dominance of non-residual Mn demonstrates an anthropogenic or external source. A more defensible interpretation is that land disturbance enhanced the mobilization of predominantly geogenic Mn from Mn-bearing bedrock and weathered materials in the upper catchment, after which the stream sediment acted as a reactive secondary reservoir. Framed this way, the sediment results become central to the overall pathway concept: the stream did not merely record Mn inputs, but functioned as a reactive secondary reservoir in which Mn was stored, redistributed, and made available for repeated redox-driven release and downstream transfer. This interpretation is compatible with the mineralogical setting of the Ilgwang district, where sulfide-bearing and hydrothermally altered granitoids provide a plausible lithogenic source of Mn that can be mobilized more efficiently when excavation and drainage modification expose fresh weathering fronts (Fletcher, 1977).

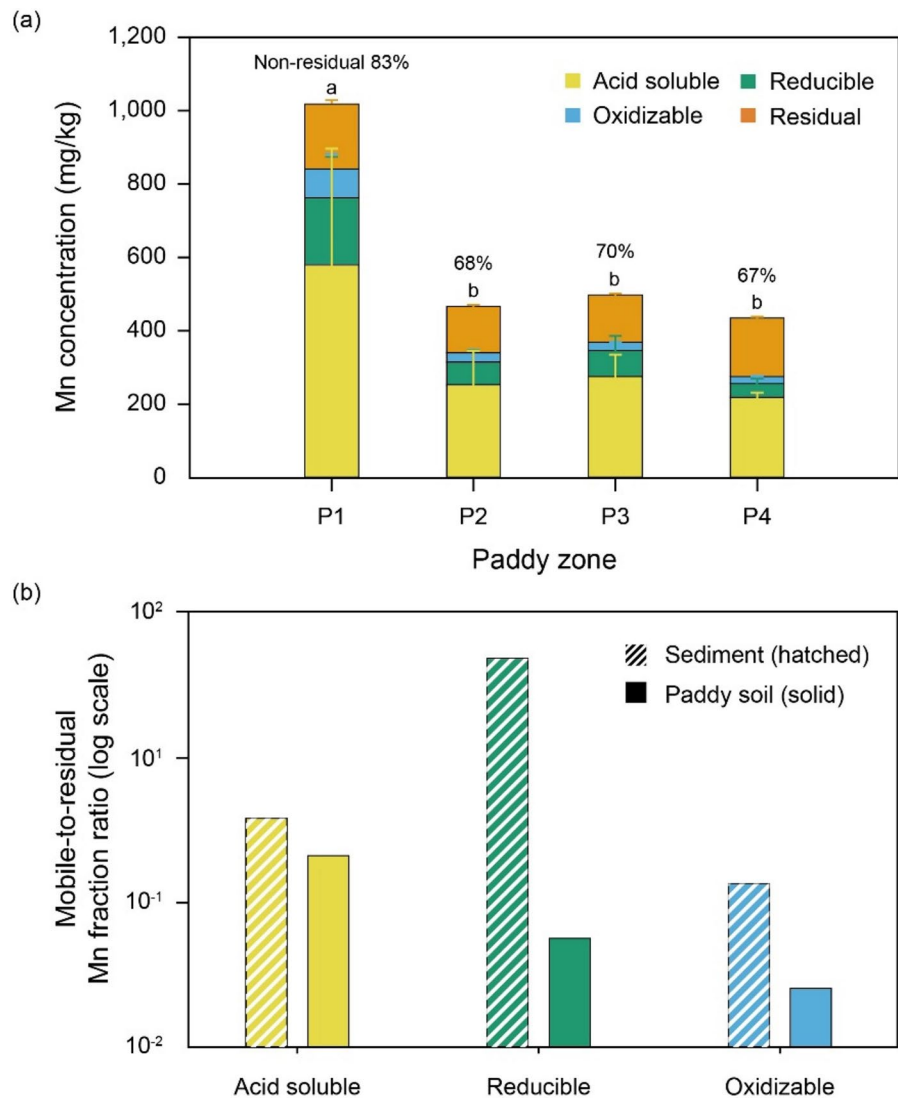
Spatial retention and redox-sensitive partitioning in paddy soils

Paddy soils showed clear Mn enrichment, with the strongest accumulation occurring in the upper irrigation zone (Fig. 4a). Total Mn concentration reached approximately 1,050 mg/kg in P1, whereas the downstream zones P2–P4 showed lower but still elevated concentrations of approximately 450–500 mg/kg. This spatial gradient closely followed the irrigation layout of the catchment, as P1 received the earliest and most direct contact with Mn-rich inflow from the adjacent stream, leading to Mn levels that were statistically significantly higher than those of all other irrigation zones (P2–P4) based on one-way ANOVA followed by Tukey's HSD test ($p < 0.001$). The paddy-soil results therefore indicate that the irrigated field system functioned as a major interception zone for Mn delivered from the upstream drainage pathway. The value of approximately 1,050 mg/kg is the field-mean across the 2021 sampling events; the harvest-stage (September) value for P1 was approximately 1,690 mg/kg and is the topsoil concentration used in the flux and inventory analysis ("[Screening-level Mn flux partitioning in the upper paddy zone](#)" section and Table 2).

The downstream decline in soil Mn is unlikely to be explained solely by dilution. Rather, it likely reflects a combination of inlet-proximal deposition, retention of Mn in the upper paddy zone, overflow-mediated redistribution, and possible redox-sensitive transformation under flooded conditions. In submerged soils, Mn oxides can be reduced to soluble Mn(II), while drainage and root-mediated oxygen release may promote re-oxidation and reprecipitation near the soil–water interface (Kirk, 2004; Ponnampetuma, 1972). The spatial patterns observed here are consistent with the redox sensitivity of Mn in flooded agricultural soils (Wang et al., 2022).

As indicated in Fig. 4a, non-residual Mn accounted for 67–83% of total Mn across the four paddy zones, indicating that a substantial proportion of soil Mn remained environmentally reactive despite the lower lability relative to stream sediment. Figure 4b directly compares Mn lability between stream sediment and paddy soil on a logarithmic scale. Although residual Mn contents were of comparable magnitude, the ratios of mobile to residual Mn were much higher in stream sediment than in paddy soil. In particular, the

Fig. 4 Mn fractionation in paddy soils and comparison of Mn lability between stream sediment and paddy soil. **a** Total Mn concentrations and operationally defined fractions (acid-soluble/exchangeable, reducible, oxidizable, and residual) in paddy soils from the four irrigation zones (P1–P4). Values are shown as mean $\pm$ SD ($n = 3$), and the percentage above each bar indicates the proportion of non-residual Mn. **b** Ratios of mobile Mn fractions to the residual fraction in stream sediment and paddy soil, illustrating substantially higher Mn lability in stream sediments than in paddy soils. Sequential extraction was performed using the modified BCR procedure. A logarithmic scale is used on the y-axis in panel (b) to accommodate the large difference in lability between stream sediment and paddy soil. Different letters indicate significant differences at $p < 0.001$



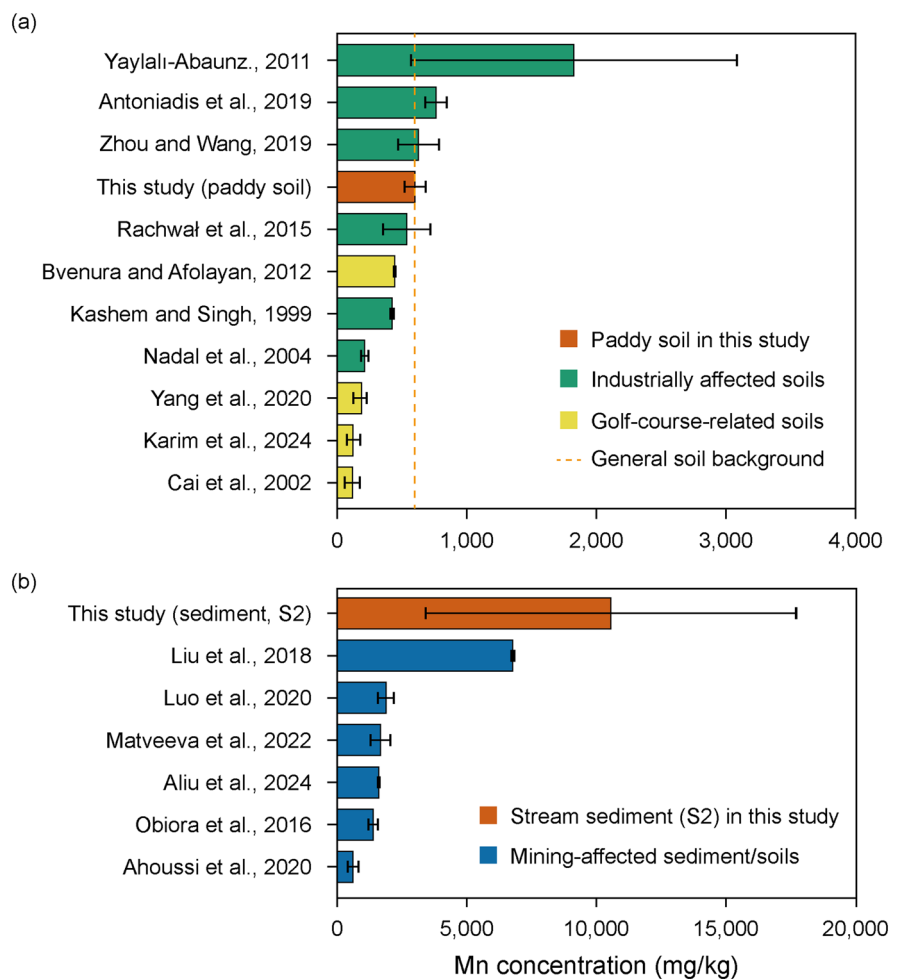
reducible/residual ratio exceeded 49.0 in sediment but remained below 0.5 in paddy soil, and none of the soil mobile/residual ratios exceeded 1.0. These results indicate that Mn became less labile after transfer from the stream into the paddy-soil matrix, suggesting partial stabilization within the soil environment. At the same time, the predominance of non-residual Mn in the soils indicates that a substantial fraction remained environmentally reactive and therefore available for remobilization under reducing conditions.

The lower mobile/residual Mn ratios observed in paddy soils may be partially attributed to preferential plant uptake. While this explanation is plausible, the current dataset precludes a quantitative differentiation

between the relative contributions of plant uptake, mineral stabilization, and seasonal redox cycling. A more comprehensive interpretation suggests that paddy soils receive labile Mn via irrigation, which is subsequently moderated by retention and transformation processes. Nonetheless, a significant redox-sensitive Mn pool is maintained, facilitating renewed mobilization during flooding. This is consistent with previous literature demonstrating that reducible Mn fractions in paddy soils are intimately linked to Mn solubility and its subsequent transfer into rice plants (Chen et al., 2019; Qin et al., 2023).

To contextualize the Mn concentrations measured in this study, values from stream sediment and paddy

Fig. 5 Contextual comparison of Mn concentrations measured in the present study with literature-reported values from industrial-, mining-, and golf-course-influenced environments. **a** Comparison of Mn concentrations in paddy soils from the present study with reported Mn concentrations in industrially affected and golf-course-related soils, together with the general soil background concentration indicated by the dashed vertical line. **b** Comparison of Mn concentrations in stream sediment from the present study at S2 with literature-reported concentrations from mining-affected sediments and soils. Values for the present study are presented as mean $\pm$ SD ($n=3$)



soil were compared with previously reported Mn concentrations from industrial, mining, and golf-course-influenced environments (Fig. 5). Reported Mn concentrations in industrially affected soils generally range from approximately 210–1,824 mg/kg (Antoniadis et al., 2019; Kashem & Singh, 1999; Nadal et al., 2004; Rachwał et al., 2015; Yaylali-Abaunz, 2011; Zhou & Wang, 2019), whereas values from mining-affected environments span a broader range of approximately 611 to 6,760 mg/kg (Ahoussi et al., 2020; Aliu et al., 2024; Liu et al., 2018; Luo et al., 2020; Matveeva et al., 2022; Obiora et al., 2016). In contrast, soils reported from golf-course-related environments generally contain markedly lower Mn concentrations, with a maximum reported value of 443 mg/kg (Bvenura & Afolayan, 2012; Cai et al., 2002; Karim et al., 2024; Yang et al., 2020). Paddy soils in the present study contained up to 1,050 mg/

kg Mn, while the mean concentration shown in Fig. 5a also exceeded the upper range reported for golf-course soils and overlapped with concentrations observed in industrially and mining-affected settings. Stream sediment from the present study showed an even higher mean Mn concentration of approximately 10,546 mg/kg, exceeding the upper range reported for many mining-affected environments (Ahoussi et al., 2020; Aliu et al., 2024; Liu et al., 2018; Luo et al., 2020; Matveeva et al., 2022; Obiora et al., 2016). These comparisons highlight the exceptional magnitude of Mn enrichment in the study catchment and are consistent with disturbance-enhanced mobilization and downstream accumulation of predominantly geogenic Mn. The unusually high Mn concentrations observed in this catchment therefore appear to reflect the convergence of three factors: a mine-influenced geogenic substrate, construction-enhanced exposure

of reactive materials, and strong hydrological connectivity between the disturbed upland source area and downstream paddies.

Transfer to rice tissues and grain accumulation

Rice tissue analysis confirmed that Mn was effectively transferred from the irrigation system and contaminated soils into the crop (Table 5). During the July growing stage, shoot Mn concentrations were generally higher than root concentrations across the four paddy zones, indicating active translocation to aboveground tissues while plants were still rapidly growing. By contrast, September samples showed a more variable tissue distribution, with root Mn concentrations exceeding shoot concentrations in some zones (e.g., P1 and P3), while other zones showed comparable or higher concentrations in shoots. This shift is consistent with a developmental change in Mn allocation as plants approached maturity, when the demand for Mn in actively photosynthesizing tissues may decline (Alejandro et al., 2020; Millaleo et al., 2010).

The most important crop result is the Mn concentration in rice grain at harvest. As shown in Table 5, mean grain Mn concentrations ranged from 90.64 to 171.47 mg/kg across P1–P4, with the highest mean value recorded at P3 (171.47 ± 21.08 mg/kg) and all zones substantially exceeding commonly reported background levels. These values greatly exceed the global background range for rice grain (1–10 mg/kg) and are also well above the 50 mg/kg level often associated with contaminated settings (Kasam et al., 2018; Pinson et al., 2015). Notably, although paddy-soil Mn was highest in the upper zone (P1), grain Mn

peaked in the intermediate zone (P3), indicating a spatial decoupling between soil Mn loading and grain accumulation that is consistent with cultivar physiology, microtopographic variation in irrigation residence time, and zone-specific redox conditions rather than with soil total Mn alone. Thus, the grain data demonstrate that the Mn pathway identified in water, sediment, and soil translated into a food-exposure pathway, even though grain Mn was not a simple linear function of soil Mn.

The physiological interpretation of these results requires caution, because the present field dataset does not directly resolve the mechanisms controlling Mn uptake and partitioning. Rice takes up Mn primarily in the divalent form, and transport systems such as OsNRAMP5 play a key role in the uptake of Mn, Fe, and Cd from soil solution (Ishimaru et al., 2012). In addition, genotype-dependent differences in grain Mn accumulation have been reported, with some cultivars accumulating substantially less Mn under otherwise similar conditions (Liu et al., 2017). These findings suggest that field-scale grain Mn concentrations in the present study likely reflect both strong environmental loading and crop-specific physiological controls, although the current dataset cannot distinguish their relative contributions.

Another issue that deserves explicit discussion is within-zone variability. Table 5 shows relatively large standard deviations in several tissues, especially for P1 grain and for roots at P1 and P4. This heterogeneity likely reflects microtopographic variation, uneven irrigation residence time, and patchy deposition of Mn-rich sediment within individual paddy zones. Therefore, while the overall source-to-crop transfer pattern is clear, the field system also appears to show

Table 5 Mn concentrations (mean $\pm$ SD, mg/kg dry weight) in rice tissues collected from four paddy zones (P1–P4). July samples include roots and shoots at the growing stage; September

samples include roots, shoots, and grain at harvest. Grain Mn was determined only for September samples

Zone	Mn concentration (mg/kg)				
	July		September		
	Root	Shoot	Root	Shoot	Grain
P1	419.22 $\pm$ 252.15	1241.50 $\pm$ 497.31	1246.73 $\pm$ 737.55	861.60 $\pm$ 79.63	124.41 $\pm$ 66.62
P2	199.26 $\pm$ 43.99	401.78 $\pm$ 37.39	890.68 $\pm$ 315.16	908.59 $\pm$ 392.98	124.29 $\pm$ 23.62
P3	343.23 $\pm$ 14.44	749.86 $\pm$ 199.04	832.21 $\pm$ 185.66	587.93 $\pm$ 54.47	171.47 $\pm$ 21.08
P4	625.43 $\pm$ 630.63	1083.41 $\pm$ 868.54	219.56 $\pm$ 120.55	615.02 $\pm$ 17.94	90.64 $\pm$ 39.80

Values are mean $\pm$ SD (n = 3) based on three representative plants collected from each paddy zone

substantial fine-scale variability in Mn exposure and plant uptake.

Screening-level Mn flux partitioning in the upper paddy zone

A screening-level Mn flux analysis was conducted for the upper paddy zone (P1) using measured irrigation-water Mn concentration, representative outlet-water Mn concentration, harvest-stage rice biomass and tissue Mn concentrations, and topsoil Mn concentration,

together with scenario-based hydrological and inventory parameters (Tables 2 and 6; Fig. 6). Figure 6 summarizes the measured and derived Mn flux terms as representative central estimates with uncertainty, allowing direct comparison among irrigation input, outlet export, plant accumulation, and apparent retained flux. The numerical values used to construct Fig. 6, including the central estimates and associated uncertainty terms, are provided in Table S3. The topsoil Mn concentration used here corresponds to the harvest-stage (September) value for

Table 6 Back-calculated mean inflow Mn concentrations required to explain the present excess Mn inventory in P1 topsoil under alternative scenarios

Reference soil Mn C_{ref} (mg/kg)	Active Mn-enriched depth, d_a (m)	2 seasons (mg/L)	3 seasons (mg/L)	4 seasons (mg/L)
500	0.05	43.9	29.9	22.9
500	0.10	85.9	57.9	43.9
500	0.15	127.8	85.9	64.9
700	0.05	36.9	25.2	19.4
700	0.10	71.8	48.5	36.9
700	0.15	106.7	71.8	54.3
1000	0.05	26.3	18.2	14.1
1000	0.10	50.6	34.4	26.3
1000	0.15	74.9	50.6	38.5

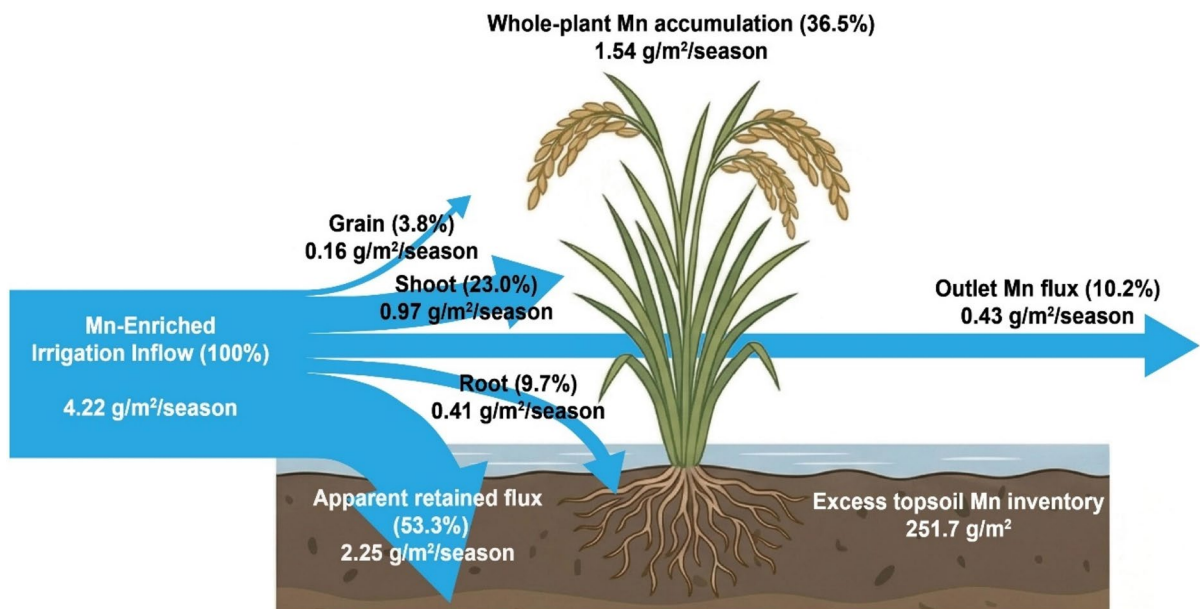


Fig. 6 Screening-level Mn partitioning in the upper paddy zone (P1). Conceptual Mn flux cycle for P1 showing representative central values for irrigation-derived Mn input, outlet

export, plant accumulation, and apparent retained flux. The excess Mn inventory in topsoil is shown separately as a standing stock rather than as a seasonal flux

P1 (approximately 1,690 mg/kg), consistent with the harvest-stage plant biomass and tissue Mn used in the same calculation; the lower field-mean of approximately 1,050 mg/kg reported in “[Spatial retention and redox-sensitive partitioning in paddy soils](#)” section reflects averaging across the 2021 sampling events.

Under the low, central, and high irrigation scenarios, the estimated Mn input flux to P1 ranged from 3.38 to 5.06 g/m²/season, equivalent to 3.02–4.52 kg/season for the entire zone. The corresponding outlet Mn flux ranged from 0.29 to 0.59 g/m²/season, or 0.26–0.53 kg/season. Harvest-stage plant Mn accumulation in P1 was estimated as 0.41 g/m² in roots, 0.97 g/m² in shoots, and 0.16 g/m² in grain, yielding a total whole-plant Mn accumulation of 1.54 g/m²/season, or 1.38 kg/season for the entire P1 zone.

Relative to the estimated Mn input, whole-plant accumulation accounted for 30.4–45.5% of incoming Mn, depending on the hydrological scenario. Grain accumulation alone represented only 3.2–4.7% of the seasonal input, indicating that only a small fraction of incoming Mn was ultimately transferred to the edible portion of the crop. Within the plant compartment, most accumulated Mn remained in roots and shoots rather than in grain. The apparent retained Mn flux, calculated as the difference between input, outlet export, and plant accumulation, ranged from 2.09 to 2.39 g/m²/season, equivalent to 1.87–2.14 kg/season across the P1 zone. This corresponded to 49.5–56.6% of the incoming Mn. Although this term does not isolate individual pathways, its magnitude indicates that a substantial fraction of irrigation-derived Mn was not recovered in outlet water or plant biomass and was therefore retained or redistributed within the soil-sediment system through processes such as sorption, sediment deposition, percolation, seepage, and lateral drainage.

For comparison with these seasonal fluxes, the excess Mn inventory in P1 topsoil was evaluated under alternative assumptions for contamination duration, active Mn-enriched soil depth, and reference soil Mn concentration. Table 6 shows that the mean inflow Mn concentration required to explain the present excess topsoil inventory varies widely across scenarios. When contamination duration was assumed to be short and Mn enrichment was assumed to extend through the full 0.15 m topsoil layer, the required mean inflow concentration substantially exceeded the values observed during the monitoring

period. By contrast, shallower active depths, longer effective loading duration, and higher reference soil Mn concentrations yielded more realistic inflow concentrations. These results indicate that the present topsoil Mn pool cannot be explained by recent dissolved Mn input alone.

Instead, the current Mn inventory in P1 topsoil is more plausibly interpreted as the cumulative product of repeated seasonal loading, earlier high-concentration episodes, partial retention of dissolved and particulate Mn, and surface-focused accumulation within the paddy system. In this sense, the upper paddy zone functions not simply as a transfer medium, but as a reactive retention compartment in which irrigation-derived Mn is partly accumulated in rice plants and partly stored or redistributed within the soil-sediment system. At the same time, the very small grain fraction confirms that rice grain is a minor mass sink despite its importance as the direct dietary exposure endpoint.

Screening-level dietary risk and management implications

The human health risk assessment indicates that rice produced in the study area may pose a substantial screening-level dietary risk (Fig. 7). Hazard quotient (HQ) values exceeded the risk threshold of 1 in children aged 0–5 years under all consumption scenarios, including occasional intake, and daily consumption resulted in HQ values above 1 for all age groups (Fig. 7a). These results identify children and frequent rice consumers as the most vulnerable exposure groups in the study population.

Comparison with previous studies further emphasizes the severity of Mn-related dietary risk in the present catchment (Fig. 7b). All four paddy zones exceeded the HQ threshold in the risk calculations, and the highest HQ value was observed at P3. Compared with the studies included in Fig. 7b, the HQ values obtained here were higher than those reported from genotype-based experimental studies (Liu et al., 2017; Pinson et al., 2015) and also higher than many values reported from industrial- or landfill-affected rice systems (e.g., Kasam et al., 2018; Shahriar et al., 2023). Rice grown in industrially affected areas of Rajshahi, Bangladesh contained 36.80–55.55 mg/kg Mn, and the corresponding EDI and HQ values exceeded the U.S. EPA reference dose, particularly

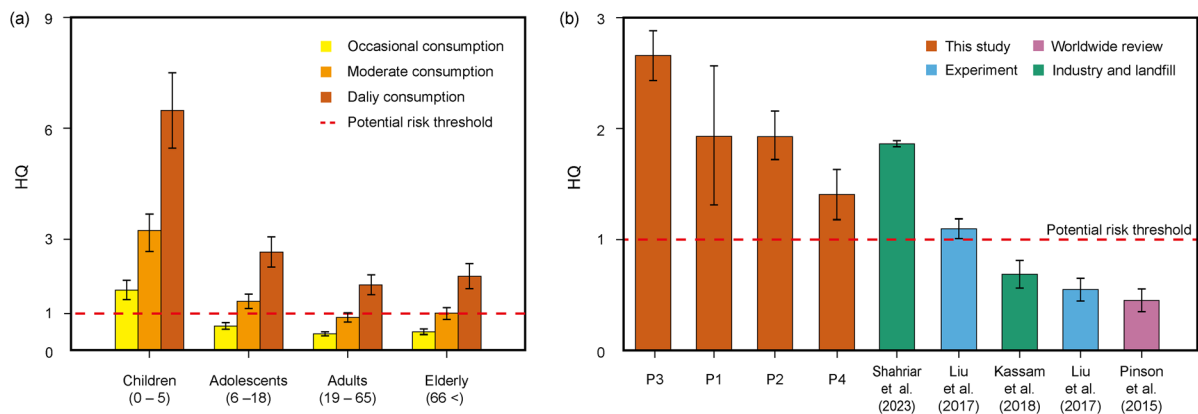


Fig. 7 Screening-level dietary risk associated with Mn in rice grain. **a** Hazard quotient (HQ) values for children, adolescents, adults, and elderly populations under occasional, moderate, and daily rice-consumption scenarios. Values are presented as mean $\pm$ SD ($n=3$). **b** Comparison of HQ values from the present study with values or concentration-based benchmarks

derived from previous studies representing contaminated field settings, genotype-based experiments, and large-scale rice surveys. The red dashed line indicates the potential non-carcinogenic risk threshold (HQ = 1). HQ values were calculated from total Mn concentrations in rice grain.

in children (Shahriar et al., 2023). Likewise, Kasam et al. (2018) reported 18.79–50.56 mg/kg Mn in rice cultivated near a landfill in Banyumas, Indonesia, and identified irrigation channels as a major contamination pathway despite relatively low soil-to-plant transfer factors. In contrast, controlled and survey-based studies generally reported lower or more variable grain Mn concentrations. Liu et al. (2017) showed that grain Mn differed substantially between rice genotypes, with approximately 60 mg/kg in PA64s and 30 mg/kg in 93–11, indicating an important genetic control on Mn accumulation. Pinson et al. (2015) further reported a wide global range of grain Mn concentrations across 1,763 rice accessions (14.35–76.00 mg/kg), with differences between flooded and unflooded cultivation conditions. Taken together, these comparisons indicate that the study catchment represents an unusually severe Mn-rice exposure setting.

The present risk evaluation is best interpreted as a screening-level assessment based on total Mn concentrations in grain rather than as a direct estimate of absorbed dose. The calculation does not account for oral bioaccessibility, local washing and cooking practices, dietary diversity, or additional Mn exposure from drinking water and other foods. These limitations do not weaken the core conclusion that rice from the study area represents a potentially important exposure pathway, but they do mean that

the reported HQ values are best interpreted as conservative upper-bound indicators rather than exact estimates of internal dose.

To test the sensitivity of these screening-level results to oral bioaccessibility, hazard quotients were rescaled by bioaccessible fractions of 0.8, 0.5, and 0.3, spanning the 20–70% reductions in rice Mn reported for washing and cooking with contaminated water (Jaafar et al., 2021). Because HQ scales linearly with the bioaccessible fraction, the qualitative conclusion was robust: under daily consumption, HQ for young children (0–5 years) remained above 1 even at the most conservative bioaccessible fraction of 0.3 (HQ $\approx$ 2.0), whereas HQ for adults fell below 1 between bioaccessible fractions of 0.5 and 0.3. The oral reference dose used here (RfD = 0.14 mg/kg-day; U.S. EPA IRIS) is itself derived from human dietary Mn studies rather than from inhalation data, so it is appropriate for dietary screening; nevertheless, it does not incorporate matrix-specific bioaccessibility, reinforcing the interpretation of the reported HQ values as conservative, total-Mn screening indicators.

The risk results also provide a natural basis for management implications. First, because the water data identify a concentrated upstream Mn source, routine screening of irrigation-source water is warranted in geologically enriched and recently disturbed catchments. Second, because the sediment data show

that the stream functions as a reactive Mn reservoir, interception or stabilization of Mn-rich sediment near the source and along deposition-prone reaches should be considered. Third, because the paddy-soil results indicate a substantial reducible Mn pool, water-management practices that alter soil redox conditions may influence Mn availability and transfer to rice. In this sense, Fig. 7 does not merely summarize risk; it highlights the need for source-pathway-receptor management across the entire irrigation system rather than at the crop stage alone.

Conclusion

This study identified a hydrologically connected Mn transfer pathway linking a disturbed upland source area to irrigation water, paddy soils, rice tissues, and screening-level dietary exposure in an agricultural catchment of southeastern Korea. Dissolved Mn was strongly enriched along the golf-course-connected drainage pathway, with concentrations remaining far above tributary-background levels and declining sharply at the paddy-entry reach during the irrigation period. This pattern indicates that the irrigated paddy system functioned as a major interception zone for Mn delivered from the upstream flow path.

Stream sediments contained exceptionally high Mn concentrations and were dominated by non-residual fractions, indicating that Mn was stored in highly reactive phases capable of repeated release under changing redox conditions. Paddy soils also showed substantial Mn enrichment, with the highest accumulation in the upper irrigation zone and non-residual Mn accounting for 67–83% of total Mn. Compared with stream sediment, Mn in paddy soils was less labile, suggesting partial stabilization after transfer into the cultivated soil matrix, while still retaining a substantial redox-sensitive pool.

A screening-level Mn flux analysis demonstrates that the upper paddy zone (P1) serves as a reactive retention compartment rather than a simple transfer medium, retaining 49.5–56.6% of irrigation-derived Mn within the soil-sediment system and sequestering 30.4–45.5% in rice biomass. While grain accumulation remains a minor mass sink (3.2–4.7%), the substantial retention of Mn indicates that P1 acts as a critical geochemical buffer that modulates downstream transport. Furthermore, back-calculation of the

topsoil inventory reveals that current Mn levels reflect cumulative historical loading rather than recent dissolved inputs alone, emphasizing the long-term role of paddy systems in storing and redistributing mine-derived contaminants.

Rice tissue analysis confirmed effective transfer of Mn from the irrigation-soil system into the crop, and grain Mn concentrations at harvest (90.64–171.47 mg/kg) greatly exceeded commonly reported background levels. Screening-level risk assessment further showed that HQ values exceeded 1 for all paddy zones, with children and frequent rice consumers representing the most vulnerable exposure groups. These results demonstrate that Mn enrichment in the study catchment was not only geochemically significant, but also directly relevant to food exposure.

These findings show that an irrigation-connected, mine-influenced catchment delivered predominantly geogenic, highly reactive Mn to paddy soils and rice grain at concentrations far exceeding background, constituting a previously overlooked dietary-exposure pathway. Whether recent land disturbance enhanced this mobilization could not be established without pre-disturbance baseline or reference-catchment data and remains a hypothesis for targeted future study. The findings nonetheless highlight the importance of source–pathway–receptor management in similar environments, including screening of irrigation-source water, control of Mn-rich sediment deposition near source-connected reaches, and water-management practices that can limit Mn remobilization in flooded paddies.

Several limitations constrain interpretation and should guide future work. The monitoring covered a single year (2021) with four campaigns, so interannual hydrological variability could not be assessed. The paddy zones lie along a single hydrological pathway and soils were composited within each zone; zone comparisons therefore represent descriptive spatial patterns rather than independently replicated contrasts and may be subject to spatial autocorrelation. A single cultivar ('Sugwang') was studied, so environmental loading and genotype-dependent uptake cannot be separated. The flux analysis is a scenario-based, dissolved-Mn screening estimate rather than a closed mass balance: outlet discharge was not gauged, and particulate Mn, seepage, and lateral flow were not quantified, so the apparent retained flux includes both

retention and unmeasured particulate input. Solid-phase partitioning is operationally defined (BCR) and was not accompanied by in situ Eh measurements, and source apportionment by mineralogical or isotopic fingerprinting was beyond the present scope. Accordingly, the geogenic-origin and redox interpretations are inferential, and the reported hazard quotients are conservative, total-Mn screening estimates. Future studies incorporating multi-year monitoring, a matched reference catchment, multi-cultivar sampling, in situ redox measurements, and geochemical fingerprinting are needed to test the inferred causal role of land disturbance and to refine the exposure estimates.

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Author contributions J.H. contributed to conceptualization, investigation, formal analysis, visualization, and wrote the original draft of the manuscript. T.K. participated in investigation and formal analysis. H.P. was involved in investigation and data curation. T.H. contributed to investigation and validation. N.J. was responsible for methodology (dietary risk assessment), writing—review & editing, and supervision. M.Y. contributed to conceptualization, supervision, funding acquisition, and writing—review & editing. All authors read and approved the final manuscript.

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Data availability The datasets generated and analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare that they have no competing financial interests or personal relationships that could have influenced the work reported in this paper.

Ethical approval Not applicable. This study did not involve human participants or animals; the dietary exposure assessment used published, aggregated national intake statistics.

Consent to participate / Consent for publication Not applicable.

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